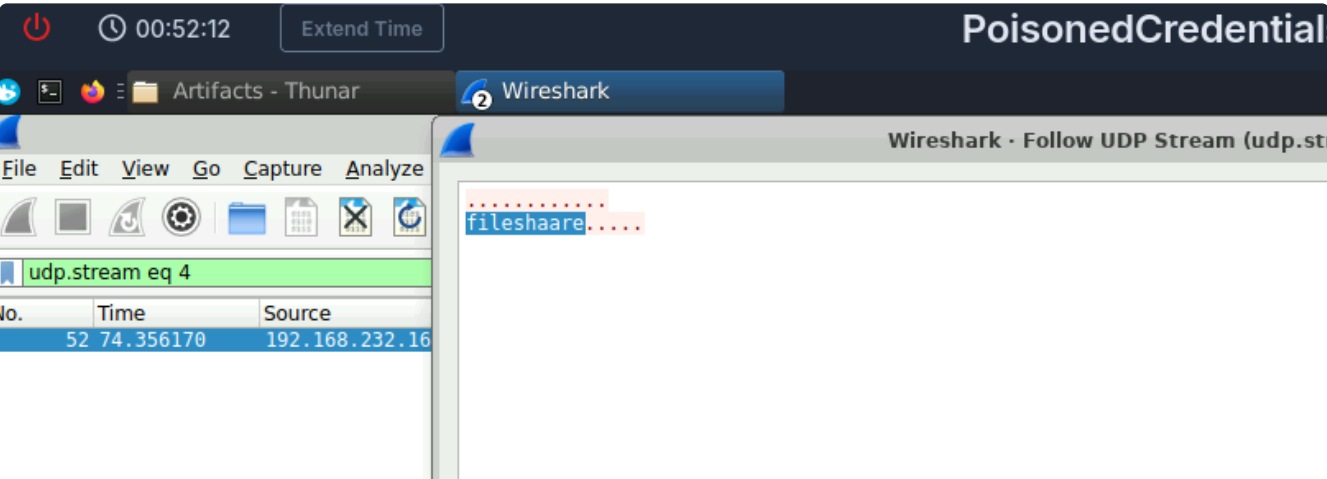


PoisonedCredentials Lab

贵组织的安全团队检测到可疑网络活动激增。他们担心您的网络可能正在遭受 LLMNR（链路本地组播名称解析）和 NBT-NS（NetBIOS 名称服务）投毒攻击。众所周知，这些攻击会利用这些协议拦截网络流量，并可能窃取用户凭据。您的任务是调查网络日志并检查捕获的网络流量。

1.在上述场景描述的事件中，攻击者利用合法机器的良性网络流量发起攻击。您能否找出 IP 地址为 192.168.232.162 的机器发出的具体错误查询语句？

LLMNR 是一种特殊 dns 协议，运行在 udp.port==5355 端口



2.我们正在调查一起网络安全事件。为了进行彻底调查，我们需要确定恶意机器的 IP 地址。请问这台恶意机器的 IP 地址是什么？

No.	Time	Source	Destination	Protocol	Length	Info
50	74.355385	192.168.232.215	192.168.232.162	MDNS	86	Standard query response 0x0000 A 192.168.232.215
54	74.359127	192.168.232.215	192.168.232.162	MDNS	98	Standard query response 0x0000 AAAA fe80::c0a9:714f:8ea7:3313
66	74.404531	192.168.232.215	192.168.232.162	MDNS	86	Standard query response 0x0000 A 192.168.232.215
68	74.406038	192.168.232.215	192.168.232.162	MDNS	98	Standard query response 0x0000 AAAA fe80::c0a9:714f:8ea7:3313
75	74.418538	192.168.232.215	192.168.232.162	MDNS	86	Standard query response 0x0000 A 192.168.232.215
78	74.420725	192.168.232.215	192.168.232.162	MDNS	98	Standard query response 0x0000 AAAA fe80::c0a9:714f:8ea7:3313
83	74.437753	192.168.232.215	192.168.232.162	MDNS	86	Standard query response 0x0000 A 192.168.232.215
86	74.439411	192.168.232.215	192.168.232.162	MDNS	98	Standard query response 0x0000 AAAA fe80::c0a9:714f:8ea7:3313

3.作为调查的一部分，识别所有受影响的机器至关重要。

第二台收到来自恶意机器的恶意响应的机器的 IP 地址是什么？

ip.src==192.168.232.215					
No.	Time	Source	Destination	Protocol	Length Info
188	254.237206	192.168.232.215	192.168.232.176	LLMNR	102 Standard query response 0x567d AAAA prin
191	254.243264	192.168.232.215	192.168.232.176	MDNS	83 Standard query response 0x0000 A 192.168.
193	254.244371	192.168.232.215	192.168.232.176	MDNS	95 Standard query response 0x0000 AAAA fe80:
195	254.247641	192.168.232.215	192.168.232.176	LLMNR	90 Standard query response 0x02c2 A prinetr
196	254.251873	192.168.232.215	192.168.232.176	LLMNR	102 Standard query response 0xb232 AAAA prin
199	254.256960	192.168.232.215	192.168.232.176	MDNS	83 Standard query response 0x0000 A 192.168.
202	254.258495	192.168.232.215	192.168.232.176	MDNS	95 Standard query response 0x0000 AAAA fe80:
203	254.260139	192.168.232.215	192.168.232.176	LLMNR	90 Standard query response 0x85e5 A prinetr
204	254.262949	192.168.232.215	192.168.232.176	LLMNR	102 Standard query response 0xd22d AAAA prin
232	398.424817	192.168.232.215	192.168.232.176	TCP	74 34643 → 445 [SYN] Seq=0 Win=64240 Len=0
234	398.425223	192.168.232.215	192.168.232.176	TCP	60 34643 → 445 [ACK] Seq=1 Ack=1 Win=64256 L
235	398.429270	192.168.232.215	192.168.232.176	SMB	127 Negotiate Protocol Request
237	398.431654	192.168.232.215	192.168.232.176	TCP	60 34643 → 445 [ACK] Seq=74 Ack=253 Win=6412
238	398.450927	192.168.232.215	192.168.232.176	SMB2	266 Negotiate Protocol Request
240	398.465476	192.168.232.215	192.168.232.176	SMB2	240 Session Setup Request. NTLMSSP NEGOTIATE

4.我们怀疑用户账户可能已被盗用。为了评估这一点，我们必须确定与被盗用账户关联的用户名。攻击者盗用的账户用户名是什么？

tcp.stream eq 11					
Packet list		Narrow & Wide		String	
No.	Time	Source	Destination	Protocol	Length Info
235	398.429270	192.168.232.215	192.168.232.176	SMB	127 Negotiate Protocol Request
236	398.431475	192.168.232.176	192.168.232.215	SMB2	306 Negotiate Protocol Response
237	398.431654	192.168.232.215	192.168.232.176	TCP	60 34643 → 445 [ACK] Seq=74 Ack=253 Win=64128 Len=0
238	398.450927	192.168.232.215	192.168.232.176	SMB2	266 Negotiate Protocol Request
239	398.451663	192.168.232.176	192.168.232.215	SMB2	388 Negotiate Protocol Response
240	398.465476	192.168.232.215	192.168.232.176	SMB2	240 Session Setup Request, NTLMSSP NEGOTIATE
241	398.466660	192.168.232.176	192.168.232.215	SMB2	453 Session Setup Response, Error: STATUS MORE PROCESSING REQUIRED, NTLMSSP CHALLENGE
242	398.476497	192.168.232.215	192.168.232.176	SMB2	598 Session Setup Request, NTLMSSP AUTH, User: cybercactus.local\janesmith
243	398.521789	192.168.232.176	192.168.232.215	TCP	60 445 → 34643 [ACK] Seq=986 Ack=1016 Win=2101248 Len=0
250	398.591702	192.168.232.176	192.168.232.215	SMB2	130 Session Setup Response

5.作为调查的一部分，我们的目标是了解攻击者的活动范围。攻击者通过 SMB 协议访问的主机名是什么？

ntlmssp.challenge.target_info

ctrl+f 搜 NetBIOS Computer Name

ntlmssp.challenge.target.info

Packet details

Narrow & Wide

Case sensitive

String

NetBIOS Computer Name

Find

Cancel

No.	Time	Source	Destination	Protocol	Length	Info
241	398.466660	192.168.232.176	192.168.232.215	SNB2	453	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP_CHALLENGE

Target Name: CYBERCACTUS

Negotiate Flags: 0xe2890235, Negotiate 56, Negotiate Key Exchange, Negotiate 128, Negotiate V

NTLM Server Challenge: cdeb5e4b7c037617

Reserved: 0000000000000000

Target Info

Length: 210

Maxlen: 210

Offset: 78

Attribute: NetBIOS domain name: CYBERCACTUS

Attribute: NetBIOS computer name: ACCOUNTINGPC

Attribute: DNS domain name: cybercactus.local

Attribute: DNS computer name: AccountingPC.cybercactus.local

Attribute: DNS tree name: cybercactus.local

Attribute: Timestamp

Attribute: End of List

Version 10.0 (Build 19041); NTLM Current Revision 15

00c0 4b 7c 03 76 17 00 00 00 00 00 00 00 00 02 00 02 K|v-----

00d0 00 4e 00 00 00 8a 00 61 4a 00 00 00 0f 43 00 59 W....a]...CY

00e0 00 42 00 45 00 52 00 43 00 41 00 43 00 54 00 55 -BERCACTU

00f0 00 53 00 02 00 16 00 43 00 59 00 42 00 45 00 52 -S...CYBER

0100 00 43 00 41 00 43 00 54 00 55 00 53 00 11 00 40 -CACTUS

0110 00 41 00 43 00 43 00 4f 00 55 00 4e 00 54 00 49 -ACCDUNT

0120 00 4e 00 47 00 50 00 43 00 04 00 22 00 63 00 79 -LGPCTcy

0130 00 62 00 65 00 72 00 63 00 61 00 63 00 74 00 75 -bercactu

0140 00 73 00 2e 00 6c 00 6f 00 63 00 61 00 6c 00 03 -s..lo cal

0150 00 3c 00 41 00 63 00 63 00 6f 00 75 00 6e 00 74 -<Account

0160 00 69 00 6e 00 67 00 50 00 43 00 2e 00 63 00 79 -lingPCcy

0170 00 62 00 65 00 72 00 63 00 61 00 63 00 74 00 75 -bercactu

0180 00 73 00 2e 00 6c 00 6f 00 63 00 61 00 6c 00 05 -s..lo cal

0190 00 22 00 63 00 79 00 62 00 65 00 72 00 63 00 61 -cyberca

01a0 00 63 00 74 00 75 00 73 00 2e 00 6c 00 6f 00 63 -ctus..lo c

01b0 00 61 00 6c 00 07 00 08 00 21 bc d0 cd 5d 04 da -a[....]..

01c0 01 00 00 00 00

Text item (text), 28 byte(s)

Packets: 269 · Displayed: 1 (0.4%)

Profile: Default